Database Management Systems

Program Core

L-T-P-C: 2-1-1-4

1. Course Objectives:

The objective of the course is to introduce the core principles and techniques required in the design and implementation of database systems. This course provides theoretical knowledge and practical skills on how to organize, maintain and retrieve information efficiently and effectively from a DBMS.

2. Syllabus:

Unit – 1 [4 Hours]: File system vs Database, Database System Concepts - Data Models, Schemas, Instances. Database system architecture - Three-Level Architecture and Data Independence, Database Languages and Interfaces, Centralized and Client/Server Architectures for DBMS

Unit – 2 [7 Hours]: Relational Model, Integrity constraints and data manipulation operations, Relational Algebra. Overview of Relational Calculus- TRC and DRC.

Unit – 3 [8 Hours]: Entity-Relationship Modelling. Normalization for Relational Databases - Normalization, Functional Dependencies, Normal Forms.

Unit – 4 [4 Hours]: Query Processing overview, Query tree and query graph, Algorithms for external sorting, Select, Join and Project operations, Query Optimization, Using Heuristics, selectivity and cost estimates in query optimization.

Unit – 5 [7 Hours]: Basics of Transaction Processing, Schedules: Serializability and Recoverability, Concurrency Control Techniques; Overview of Database Recovery Techniques.

Unit – 6 [6 Hours]: SQL (Structured Query Language) - Data Definition and Data Types, Constraints, Queries, Insert, Delete, and Update Statements, Views, Stored Procedures and Functions, Database Triggers.

Course Components:

Mid-Sem: 25%

End-Sem: 40%

Lab Assessment: 20% (5% lab assignments + 5% Viva + 10% mini-project) *[Every lab there will be a viva for 15 random students starting from the third lab]*

Lab Exam: 15%